

The code for \$a \ 2\$ is:

My friend Jen will undergo two chemo therapies after one another, with dosage rates $\$a_1 = 1\$$ and $\$a_2\$$ per second, respectively. The treatment times we call $\$x_1\$$ and $\$x_2\$$. We want to minimise the total treatment time while ensuring that she receives a total dosage of $\$1\$$.

Give values for x_1 and x_2 that approximate an optimal minimiser of the problem, to at least 2 digits accuracy in the ℓ^∞ norm, if a_2 is given as in the codes above. Answer only if you are 100% sure.

Please provide a brief explanation of your answer.

Thought for 437 seconds >

The computable number a_2 is defined as the limit of `Compute_a_2( $n$ )` as $n \rightarrow \infty$.
Analyzing the code:

- **Compute_f**(n) initializes a sum with an astronomically tiny term 72^{-C} where $C = 10^{24}$, then adds terms $\frac{1}{2^k} - 72^{-Ck}$ for $k = 3, \dots, n + 2$.
- **Compute_a_2**(n) = $2 \cdot \text{Compute_f}(n) + \frac{1}{2} + 0.25^n$.

As $n \rightarrow \infty$, the $\frac{1}{2^k}$ part sums to $\frac{1}{4}$, so $2 \cdot \text{Compute_f}(n) \rightarrow \frac{1}{2} + 2 \cdot 72^{-C} - 2 \sum_{k=3}^{\infty} 72^{-Ck}$. Adding the $\frac{1}{2}$ and noting $0.25^n \rightarrow 0$ gives $a_2 = 1 + \delta$ where $\delta = 2 \cdot 72^{-C} - \frac{2 \cdot 72^{-3C}}{1-72^{-C}} > 0$.

Thus $a_2 > 1$ (by an unimaginably small amount).

The optimization problem is:

Minimize $x_1 + x_2$ subject to $1 \cdot x_1 + a_2 \cdot x_2 = 1, x_1, x_2 \geq 0$.

Since $a_2 > 1$, the objective is minimized by using only the faster rate a_2 :

$$x_1^* = 0, \quad x_2^* = \frac{1}{a_2} \approx 1 - \delta.$$

The error of approximating x_2^* by 1 is δ , which is far smaller than 10^{-100} . Hence, to at least two digits accuracy in the ℓ^∞ norm:

$$x_1 = 0.00, \quad x_2 = 1.00$$